MIL-D-13383 (Ord)
19 April 1954

MILITARY SPECIFICATION

DI-(2-ETHYLHEXYL)-ADIPATE

1. SCOPE

1.1 Scope.- This specification covers one type of di-(2-ethylhexyl)-adipate for use in polyisobutylene binder, used in explosive compositions.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
VV-L-791 - Lubricants, Liquid Fuels, and Related Products, Methods of Inspection, Sampling, and Testing.

MILITARY
MIL-G-2550 - General Specification for Ammunition, Except Small Arms Ammunition.

STANDARDS

MILITARY
MIL-STD-129 - Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE
Consolidated Freight Classification - Ratings, Rules and Regulations.

(Application for copies of this publication should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Sta.,Chicago 6, Ill.)

AMERICAN TRUCKING ASSOCIATIONS, INC.
National Motor Freight Classification Rules.

(Application for copies of this publication should be addressed to the American Trucking Associations, Inc., 1426-16th St.,N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Physical and chemical requirements.- The material shall conform to the following physical and chemical requirements, compliance with which shall be determined in accordance with the corresponding test procedure contained in section 4 of this specification:

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Property	Required value
Color	Not darker than the color standard (see 4.4.1)
Specific gravity at 20/20°C.	0.927 ± 0.002
Refractive index at 20°C.	1.4470 ± 0.0010
Flash point, °F, minimum	385
Fire point, °F, minimum	420
Acid number, maximum	0.07
Viscosity at 100°F, centistokes	8.46 ± 0.30
Saponification number	302 ± 2

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- Unless otherwise specified, a lot shall consist of 50,000 pounds, maximum. However, when shipment is made in tank cars, the contents of each tank car shall constitute a lot.

4.2 Sampling.

4.2.1 Containers.- Ten percent of the containers in a lot, but in no case more than 10 or less than 3 containers, shall be selected by the inspector so as to be representative of the lot. If there are less than 3 containers in the lot, all the containers shall be sampled. The material in each container shall be mixed to a uniform consistency throughout, and approximately 8 ounces taken from each container. Each primary sample shall be placed in an airtight glass bottle, having a glass stopper or a cap containing a cork or cardboard liner covered with metal foil, and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight glass bottle, having a glass stopper or a cap containing a cork or cardboard liner covered with metal foil, and labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and lot size. All the acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of this specification. All primary samples shall be held for possible future examination should the composite sample fail to meet the requirements.

4.2.2 Tank cars.- By means of a clean, weighted, small-neck glass bottle, approximately 1 quart of the material shall be removed. The unstoppered bottle shall be rinsed with the material being sampled, lowered to the bottom of the tank by means of a cord, chain, or rod, and immediately withdrawn to the surface. In order to obtain a representative cross-section of the material, the speed of lowering and raising the bottle shall be constant and so regulated that ;the bottle is just filled as it reaches the surface of the liquid. The sample shall be placed in an airtight glass bottle, having a glass stopper or a cap containing a cork or cardboard liner covered with metal foil, and labeled to show the name of the material, manufacturer,

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plant, contract or order number, lot number, and lot size. All acceptance tests for the tank car lot shall be made on this sample.

4.3 Inspection.- Inspection shall be as specified in Specification MIL-G-2550.

4.4 Tests.

4.4.1 Color.

4.4.1.1 Standard color solution.- Transfer 1.00 ± 0.01 gram of C.P. iodine and 5.0 ± 0.1 gram of C.P. potassium iodide to a 150-ml. beaker. Add approximately 25-ml. of distilled water to dissolve the iodine and the potassium iodide. Transfer the solution quantitatively to a 1,000-ml. volumetric flask, dilute to the mark with distilled water, and mix thoroughly. The standard solution shall not be used after 24 hours.

4.4.1.2 Procedure.- Fill a clean 50-ml. Nessler tube to the mark with the sample. Fill another clean 50-ml. Nessler tube to the mark with the prepared standard color solution. Hold the two tubes side by side, and compare their colors using transmitted light. Consider the color of the sample to be satisfactory if it is not darker than that of the standard.

4.4.2 Specific gravity.- Determine the specific gravity of the sample at 20/20° C by means of any suitable pycnometer.

4.4.3 Refractive index.- Determine the refractive index at 20°C using any Abbé refractometer, or any other equally accurate instrument.

4.4.4 Flash point.- The flash point shall be determined in accordance with method 1103 of Specification VV-L-791.

4.4.5 Fire point.- The fire point shall be determined in accordance with method 1103 of Specification VV-L-791.

4.4.6 Acid number.- Transfer approximately 100 ml. of a solution, consisting of equal parts, by volume, of 95-percent alcohol and reagent-grade benzene, to a 300-ml. Erlenmeyer flask. Heat the flask and contents to incipient boiling on a steambath. Add 2 or 3 drops of approximately 1-percent phenolphthalein indicator to the flask contents and titrate with approximately 0.01N potassium hydroxide solution, to a faint pink color. Add 50 ml. of sample, and again titrate the contents of the flask to a faint pink color. Calculate the acid number (milligrams of potassium hydroxide required to neutralize 1 gram of the sample), as follows:

$$\text{Acid number} = \frac{56.1\,(AN)}{BD}$$

where:

A = ml. of potassium hydroxide required to titrate the sample.
N = normality of potassium hydroxide solution.
B = volume of sample titrated.
D = specific gravity of the material (see 4.4.2).

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4.4.7 Viscosity.- The viscosity shall be determined in accordance with method 305 of Specification VV-L-791, using a calibrated Ubbelohde viscosimeter.

4.4.8 Saponification number.- Transfer an accurately weighed portion of approximately 2 grams of the sample to a 300-ml. ground-glass Erlenmeyer or round-bottom flask. Add 50 ml. of approximately 0.5N alcoholic potassium hydroxide solution. Connect the flask to an air condenser, approximately 250 ml. in length, and reflux for 3 hours on a steam bath at such a rate that the major portion of the alcohol will condense in the lower half of the condenser as indicated by a sharp drop in temperature in the lower 125 mm. of the condenser. Allow the flask and contents to cool to room temperature. Remove the condenser and rinse with three 10-ml. portions of neutralized 95-percent alcohol, collecting the washings in the flask. Add 2 or 3 drops of 1-percent phenolphthalein indicator to the flask contents, and titrate with approximately 0.5N hydrochloric acid to the first permanent faint pink color. Make a blank determination on 50 ml. of the alcoholic KOH solution in accordance with the above procedure. Calculate the saponification number as follows:

$$\text{Saponification number} = \frac{56.1\ (A-B)\ N}{W}$$

where;

A = ml. of hydrochloric acid required to titrate the sample.
B = ml. of hydrochloric acid required for blank titration.
N = normality of hydrochloric acid solution.
W = weight of sample.

4.5 Resubmission of retest.- If the composite sample, or any primary sample subjected to test, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. The resubmitted lot may be accepted, provided new samples, selected in accordance with 4.2, pass all the tests required by this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Unless otherwise specified the material shall be packed in shipping containers conforming to the Consolidated Freight Classification Rules, for railroad shipments, or to the National Motor Freight Classification Rules, for truck shipments, as applicable to the mode of transportation used.

5.2 Marking.- The shipping containers shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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